

Calculus I Lesson 08

Theorem: a) $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ b) $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0$

Note: $\lim_{x \rightarrow 0} \frac{\cos x - 1}{x} = \lim_{x \rightarrow 0} \frac{-1(1 - \cos x)}{x} = -1 \cdot \lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0$

Note: $\lim_{x \rightarrow 0} \frac{x}{\sin x} = \lim_{x \rightarrow 0} \frac{1}{(1/x)} \cdot \frac{1}{\sin x} = \lim_{x \rightarrow 0} \frac{1}{\frac{\sin x}{x}} = \frac{\lim_{x \rightarrow 0} 1}{\lim_{x \rightarrow 0} \frac{\sin x}{x}} = \frac{1}{1} = 1$

1) Find $\lim_{x \rightarrow 0} \frac{\sin x}{3x} = \underline{\hspace{2cm}} ?$

Hint: $\frac{\sin x}{3x} = \frac{1}{3} \cdot \frac{\sin x}{x}$

2) Find $\lim_{x \rightarrow 0} \frac{\sin^2 x}{5x} = \underline{\hspace{2cm}} ?$

Hint: $\frac{\sin^2 x}{5x} = \frac{1}{5} \cdot \frac{\sin x}{x} \cdot \sin x$

3) Find $\lim_{x \rightarrow 0} \frac{6x}{\sin 4x} = \underline{\quad ? \quad}$

Hint: $\frac{6x}{\sin 4x} = \frac{6 \cdot \frac{1}{4} \cdot 4x}{\sin 4x} = \frac{3}{2} \cdot \frac{4x}{\sin 4x}$

Note: $\lim_{x \rightarrow 0} \frac{x}{\sin x} = 1$ and $\lim_{x \rightarrow 0} \frac{4x - 1}{\sin(4x)} = 1$

4) Find $\lim_{x \rightarrow 0} \frac{\sin(-3x)}{4x} = \underline{\hspace{2cm}} ?$

Hint: $\frac{\sin(-3x)}{4x} = \frac{-\sin(3x)}{4 \cdot \frac{3}{3} \cdot x} = \frac{-\sin(3x)}{\frac{4}{3} \cdot 3x} = -\frac{3}{4} \cdot \frac{\sin(3x)}{3x}$

Note: $\lim_{x \rightarrow 0} \frac{\cos x - 1}{x} = 0$ and $\lim_{x \rightarrow 0} \frac{\cos 3x - 1}{3x} = 0$

5) Find $\lim_{x \rightarrow 0} \frac{\sin x + 5x}{3x} = \underline{\hspace{2cm}} ?$

Hint: $\frac{\sin x + 5x}{3x} = \frac{\sin x}{3x} + \frac{5x}{3x} = \frac{1}{3} \cdot \frac{\sin x}{x} + \frac{5}{3}$

6) Find $\lim_{x \rightarrow 0} \frac{\sin^2 x + 7x}{3x} = \underline{\hspace{2cm}} ?$

Hint: $\frac{\sin^2 x + 7x}{3x} = \frac{\sin^2 x}{3x} + \frac{7x}{3x} = \frac{1}{3} \cdot \frac{\sin x}{x} \cdot \sin x + \frac{7}{3}$

7) Find $\lim_{x \rightarrow 0} \frac{1 - \cos x}{5x} = \underline{\hspace{2cm}} ?$

8) Find $\lim_{x \rightarrow 0} \frac{1 - \cos 3x}{5x} = \underline{\hspace{2cm}} ?$

Hint: $\frac{1 - \cos 3x}{5x} = \frac{1 - \cos 3x}{5 \cdot \frac{3}{3}x} = \frac{1 - \cos 3x}{\frac{5}{3} \cdot 3x} = \frac{3}{5} \cdot \frac{1 - \cos 3x}{3x}$

Note: $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0$ and $\lim_{x \rightarrow 0} \frac{1 - \cos 3x}{3x}$

9) Find $\lim_{x \rightarrow 0} \frac{4x + 1 - \cos x}{5x} = \underline{\hspace{2cm}} ?$

Hint: $\frac{4x + 1 - \cos x}{5x} = \frac{4x}{5x} + \frac{1 - \cos x}{5x} = \frac{4}{5} + \frac{1}{5} \cdot \frac{1 - \cos x}{x}$

10) Find $\lim_{x \rightarrow 0} \frac{x^2 \cdot \cos x - x^3}{5x^2} = \underline{\hspace{2cm}} ?$

Hint: $\frac{x^2 \cdot \cos x - x^3}{5x^2} = \frac{x^2 \cdot (\cos x - x)}{5x^2} = \frac{1}{5} \cdot (\cos x - x)$

11) Find $\lim_{x \rightarrow 0} \frac{\sin x}{1 - \cos x} = \underline{\hspace{2cm}} ?$

Hint:
$$\frac{\sin x}{1 - \cos x} \cdot \frac{1 + \cos x}{1 + \cos x} = \frac{\sin x + \sin x \cdot \cos x}{1 - \cos^2 x} = \frac{\sin x + \sin x \cos x}{\sin^2 x}$$

$$= \frac{\sin x}{\sin^2 x} + \frac{\sin x \cos x}{\sin^2 x} = \frac{1}{\sin x} + \frac{\cos x}{\sin x} = \frac{1}{\sin x} + \cot x$$

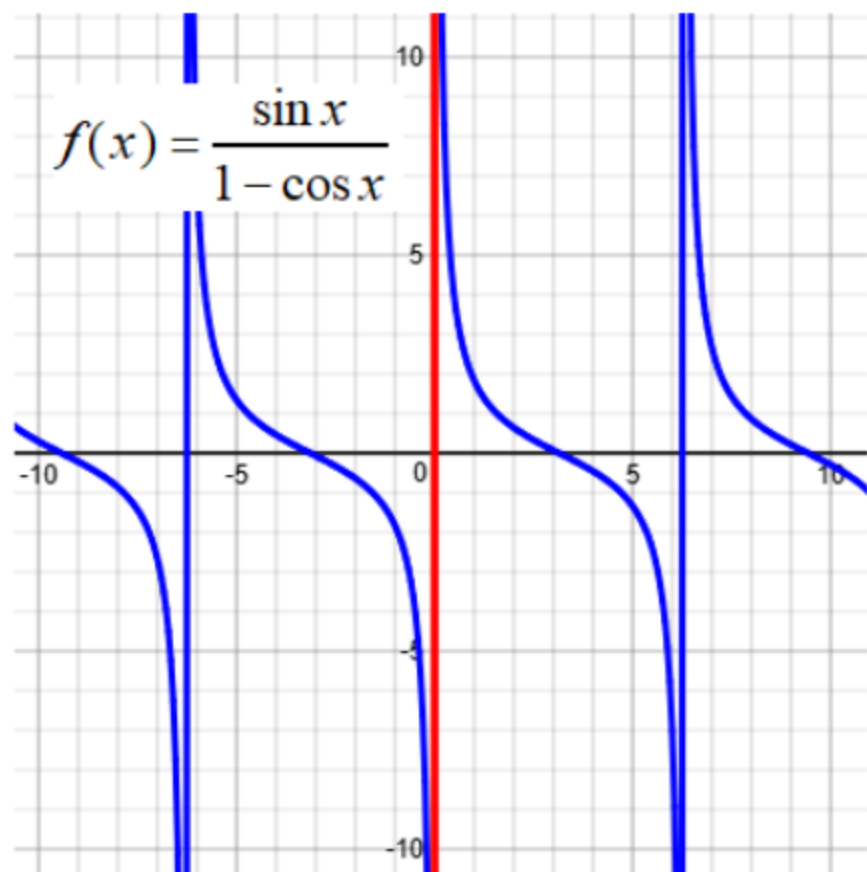
$\lim_{x \rightarrow 0} \frac{\sin x}{1 - \cos x} = \underline{\hspace{2cm}} ?$

Let $f(x) = \frac{\sin x}{1 - \cos x}$

$\lim_{x \rightarrow 0^-} f(x) = \underline{\hspace{2cm}} ?$ limit of $f(x)$ as x approaches 0 from the left

$\lim_{x \rightarrow 0^+} f(x) = \underline{\hspace{2cm}} ?$ limit of $f(x)$ as x approaches 0 from the right

$\lim_{x \rightarrow 0} f(x) = \underline{\hspace{2cm}} ?$ limit of $f(x)$ as x approaches 0



12) Find $\lim_{x \rightarrow 0} \frac{\cos x}{1 - \sin x} = \underline{\hspace{2cm}} ?$

Hint: $\frac{\cos x}{1 - \sin x} \cdot \frac{1 + \sin x}{1 + \sin x} = \frac{\cos x + \sin x \cdot \cos x}{1 - \sin^2 x} = \frac{\cos x + \sin x \cdot \cos x}{\cos^2 x}$

$$= \frac{\cos x}{\cos^2 x} + \frac{\sin x \cdot \cos x}{\cos^2 x} = \frac{1}{\cos x} + \frac{\sin x}{\cos x} = \frac{1}{\cos x} + \tan x$$

$\lim_{x \rightarrow 0} \frac{\cos x}{1 - \sin x} = \underline{\hspace{2cm}} ?$

Let $f(x) = \frac{\cos x}{1 - \sin x}$

$\lim_{x \rightarrow 0^-} f(x) = \underline{\hspace{2cm}} ?$

limit of $f(x)$ as x approaches 0 from the left

$\lim_{x \rightarrow 0^+} f(x) = \underline{\hspace{2cm}} ?$

limit of $f(x)$ as x approaches 0 from the right

$\lim_{x \rightarrow 0} f(x) = \underline{\hspace{2cm}} ?$

limit of $f(x)$ as x approaches 0

